

B Feature Analysis

Figure 4 through Figure 8 present leave-one-out feature importance, $\Delta\hat{m}_{s^t}$ represents the difference between the baseline of using all features on the $\hat{m}_s^t$ across all timesteps. Features are grouped into three categories: **Static**, **CDCL-based**, and **BnB-based**. Positive $\Delta\hat{m}_{s^t}$ values indicate a decrease in predictive performance when the feature is removed, representing higher importance.

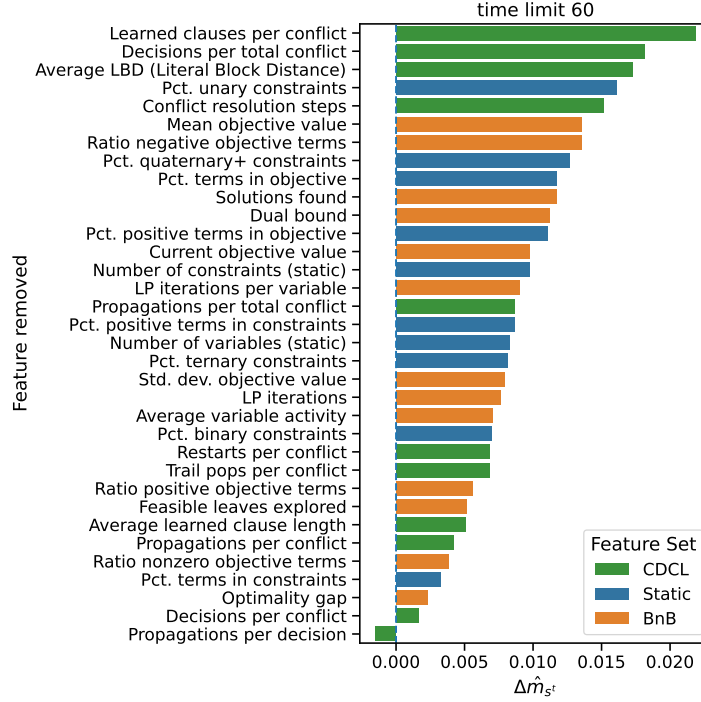


Fig. 4. Feature importance at timestep 60. CDCL-based features dominate early predictions.

At the early stage (time limit of 60 seconds), predictive performance is primarily influenced by CDCL-based features. Metrics related to learned clauses, decisions per conflict, average LBD, and conflict resolution steps exhibit the highest $\Delta\hat{m}_{s^t}$ values, indicating that early predictions are strongly dependent on clause-learning dynamics and conflict structures. BnB features such as the mean objective value and the number of solutions found have a moderate influence, while static structural features contribute only minimally. This pattern suggests that early performance estimation relies heavily on conflict-driven solver behavior rather than global structural or objective-related properties.

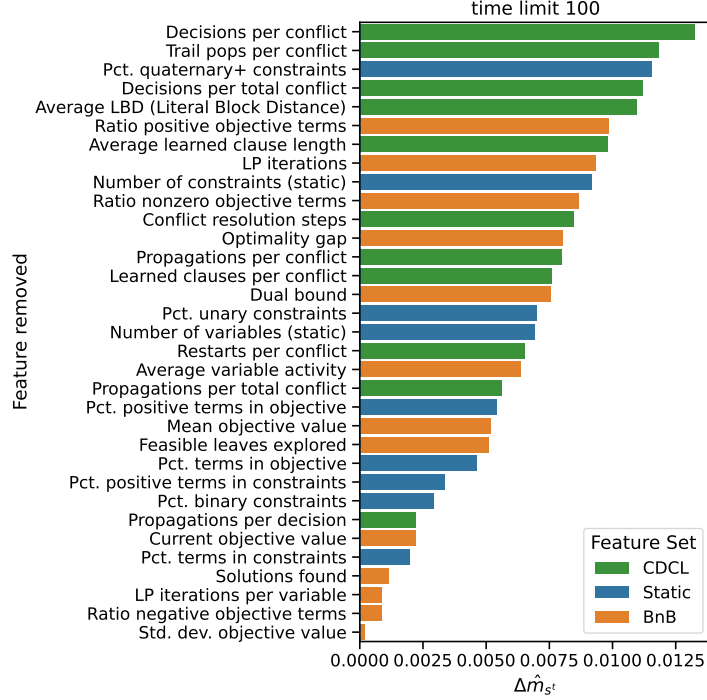


Fig. 5. Feature importance at timestep 100. CDCL features remain dominant, with increasing structural and BnB influence.

By timestep 100, CDCL features remain dominant but the difference relative to other feature groups narrows. Decisions per conflict and trail pops per conflict continue to rank highly, while structural constraint features, such as the percentage of quaternary constraints, gain importance. BnB-related metrics, including the optimality gap and ratios of positive objective terms, also become more relevant. Static features, such as the number of variables and constraints, increase their contribution compared to the earliest phase. Overall, the model begins integrating both search behavior and structural composition, indicating that conflict activity remains central but objective and constraint structures start to play a more significant role in predictive performance.

At the mid stage (timestep 300), feature importance becomes more evenly distributed across all categories. Static features, including the percentage of terms in constraints and the binary constraint ratio, rise in prominence. BnB-related metrics, such as feasible leaves explored and objective-related features, also gain importance. CDCL-based features remain relevant but no longer dominate the predictions. Some features display slightly negative $\Delta\hat{m}_{st}$ values. This shift indicates that solver behavior depends less exclusively on clause-learning dynamics and more on global structural and objective characteristics.

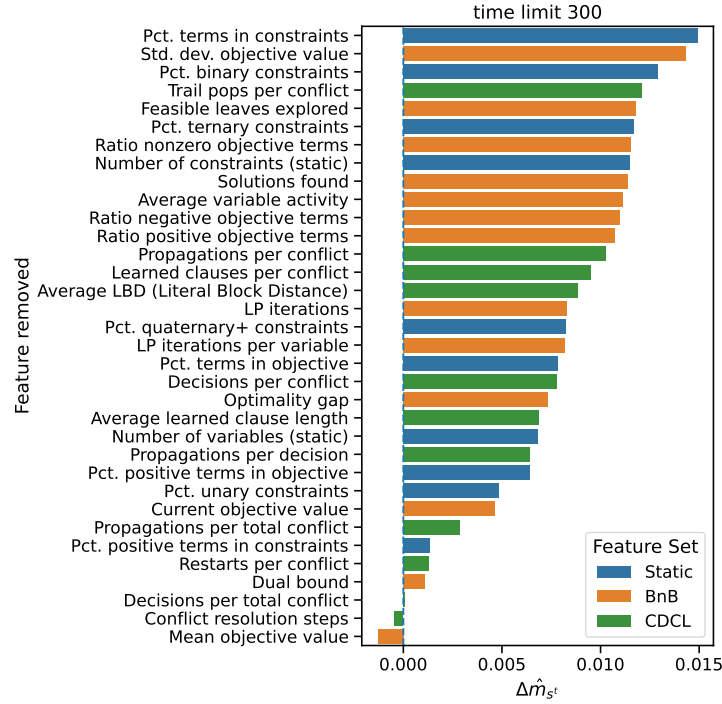


Fig. 6. Feature importance at timestep 300. Importance is more evenly distributed across feature categories.

During the late-mid stage (timestep 600), BnB features dominate the importance rankings. Metrics related to LP iterations per variable, the percentage of terms in constraints, and standard deviation of objective values rise to the top, while CDCL features remain present but are no longer leading. Static structural features maintain moderate influence, highlighting the solver’s shift toward branch-and-bound behavior. Predictive performance is now primarily driven by optimization-specific dynamics rather than conflict-learning.

By the final stage (timestep 3600), static features become more prominent. The most important predictors include the percentage of positive terms in constraints and in the objective, the ratio of binary constraints, average variable activity, propagations per conflict, and the total number of variables, while objective-related and other Branch-and-Bound features remain relevant. This indicates that the solver has reached a more stable phase in which performance is shaped primarily by structural properties, whereas probing-based features that reflect early search behavior become less informative over time.

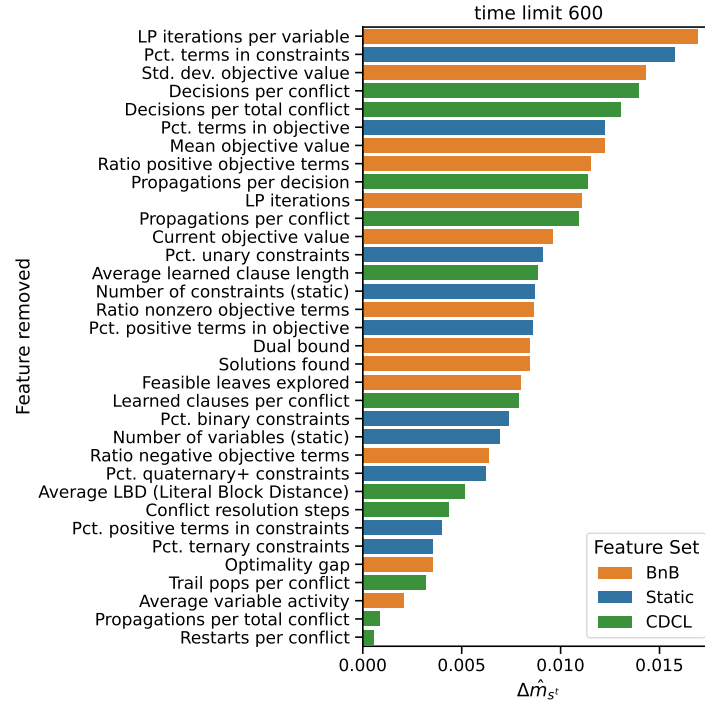


Fig. 7. Feature importance at timestep 600. BnB features dominate, reflecting optimization-driven solver behavior.

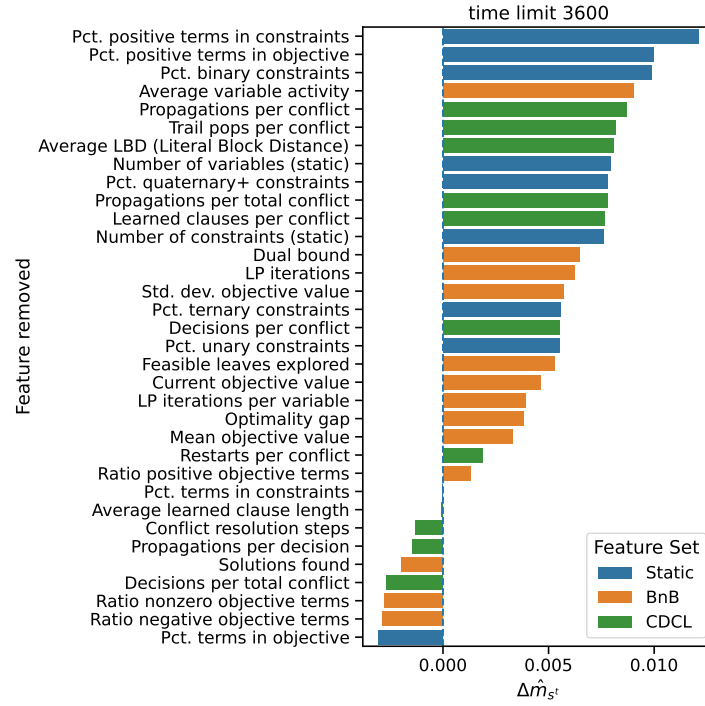


Fig. 8. Feature importance at timestep 3600. Static features dominate long-term predictions; probing features are less influential.